More valuable materials are to be produced from those of a lesser value. New substances can be produced by nuclear reactions in very small amounts only. Therefore, their production is worth while only when they happen to be especially valuable. These extremely precious materials, which represent a considerable value, even in small quantities, are the radioactive materials. Their value consists in their radiation, which can be utilized in many ways and is considerable even when the quantity of the substance is very small. For this reason, the most important application of nuclear physics at the present time consists in an artificial production of radioactive materials.

Radioactive substances can be used for various purposes. They have been used, in medicine, for several decades, to irradiate malignant tumours, which experience has proved to be far more seriously injured by radioactive emissions than healthy tissue. In the majority of cases, of course, x-rays are used for this purpose. But whenever a complication arises that makes it difficult to approach the diseased spot without injuring other tissue, the use of radioactive preparations is preferred. However, since the natural radioactive substances are available for medical purposes in very limited quantities only —apart from radium itself, only *mesothorium,* discovered by O. Hahn, comes into consideration at all—it is hoped to achieve important progress in medical research by the production of artificial radioactive materials in sizable amounts, especially those which possess chemical properties different from those encountered in the natural ones.

Radioactive substances have still another use: Quite minute quantities of them are mixed with luminescent materials, which give out a constant luminescence due to the effect of radiation. Their best known application is in the luminous dials and hands of watches.

The radiation of radioactive preparations is used also for the testing of industrial materials for internal, structural imperfections, to which use x-rays are customarily put. The gamma radiation is the one utilized for this purpose. This procedure is applied especially to pieces too thick to be penetrated by x-rays, but which will still let gamma radiation pass through. This method of testing industrial materials has the great advantage over other procedures that it does not injure the material itself.

Let us now discuss the production and utilization of radioactive substances in a little more detail.

In order to produce artificial radioactive substances, a suitable substance is irradiated by neutrons in a uranium reactor, or by protons or deuterons in a high-voltage generator or cyclotron. A certain practical difficulty is encountered here, owing to the fact that the substance which it is desired to produce is often present in the original material in the form of minute, imponderable impurities. This substance may be chemically different from the original substance or it may be identical with it, that is to say one of its own